[redacted]

From: Steve Greer [redacted]>
Sent: Saturday, April 20, 2013 3:43 AM
To: [redacted]
Subject: Fwd: confidential shared....

p and s...steve

---------- Forwarded message ----------
From: [redacted]>
Date: Fri, Apr 19, 2013 at 11:40 PM
Subject: confidential shared....
To: "Steven M. Greer [redacted]

A potential candidate:

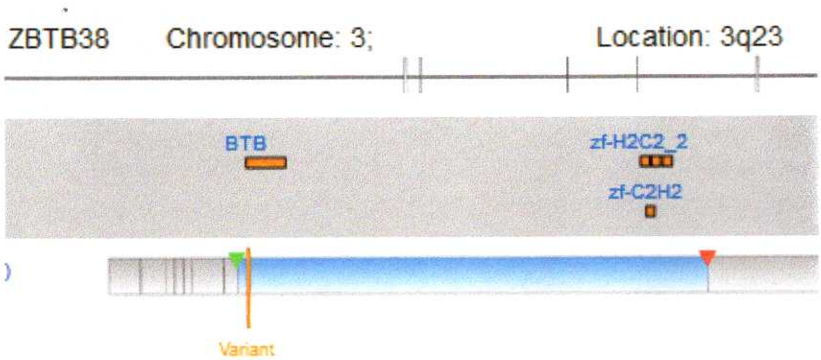
The genome looks good (filter impacts and metrics like Ti/Tv are normal) and this ZBTB38 variant surely looks relevant:

Chr...	Position	Referen...	Sample ...	Gene Region	Gene Symbol	Transcript ID	Transcript Variant	Protein Variant
3	141161303	G	T	Exonic	ZBTB38	NM_001080412.2	c.73G>T	p.E25*

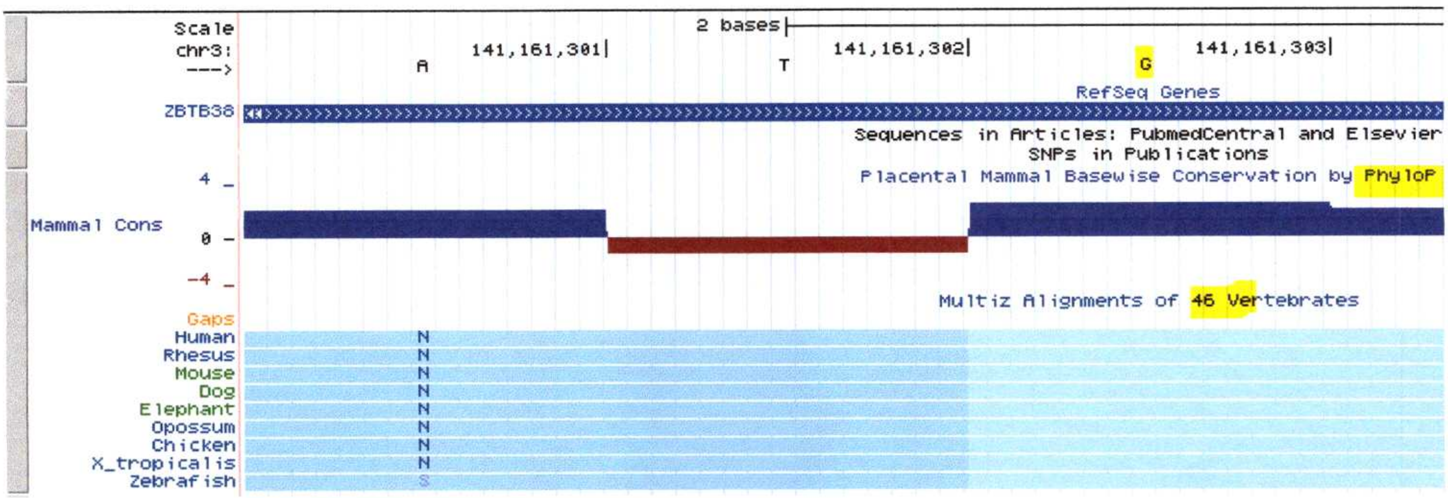
Since:

1. GWAS studies have consistently found (common) variants around this genes have the strongest contribution to height and across multiple ethnicities seem to always have strongest association with short stature (and a tie-in to my favorite NF-kappaB—a gene I cloned years ago that launched my career at Stanford): http://www.ncbi.nlm.nih.gov/pubmed/23596138 is on of most recent studies. For functional characterizes ZBTB38, see: http://www.ncbi.nlm.nih.gov/pubmed/16354688)

2. Stop codon gain at the beginning of protein (complete truncation) at front of dimerization (BTB) domain (http://www.ncbi.nlm.nih.gov/Structure/cdd/cddsrv.cgi?uid=cl02518)



3. Residue affected is exceptionally highlight conserved across vertebrates:



(from http://genome.ucsc.edu/cgi-bin/hgTracks?hgHubConnect.destUrl=..%2Fcgi-bin%2FhgTracks&clade=mammal&org=Human&db=hg19&position=chr3%3A141161303&hgt.positionInput=chr3%3A141161303&hgt.suggestTrack=knownGene&Submit=submit&pix=1024)

4. Never observed in >6000 normal genomes (Exome Server Project covers this region and has no variant reports at this location, no dbSNP record so likely has never been observed in other sequences)

5. Heterozygous genotype is consistent with dominant de novo (presumed genetic model): lack of observation of this variant in any genome to date means likely embryonic lethal if homozygous, so heterozygous genotype is consistent with presumed dominant de novo effect (presumably nobody with this variant or similar one in the gene could reproduce)

6. The sample read depth is decent (19) and call quality is good (35) -- I would recommend amplifying this region and Sanger sequencing to confirm however, since I'm not sure what upstream pipeline was used to call the variants and there is some risk this could be mis-called given those call confidence metrics.

7. Could try making an alien mouse next ☺ -- surprisingly there are no transgene/KO phenotypes from Jax yet (though is also broadly expressed in mice): http://www.informatics.jax.org/searches/allele_report.cgi?_Marker_key=84053

This work was largely enabled/done by the Founder of a genomics analysis company I indirectly helped start years and years ago. At this point the results will all be "correlative"-- this gene is thought to do this or that (stature, bone, etc.). The "proof" is in knocking the gene out in mice and seeing if a similar phenotype occurs. The problem is there can be all kinds of counterbalancing mutations that stop the lethality. So, we could make a mouse model and it would die in gestation because we don't know which of the several hundred thousand other mutations might be required to allow survival.

Bottom line at this stage: this is as close to a smoking gun for what might cause this. But if I wanted to put on my (I believe in aliens) skeptic hat… one could always formally argue this is part of a DESIGNED genome.

To be continued…

Getting on the plane shortly.

[redacted]

[redacted]

From: Steve Greer <[redacted]
Sent: Thursday, April 18, 2013 11:58 AM
To: [redacted]
Subject: Fwd:

fyi
[redacted]ave link to article [redacted]s referencing - I will refer to it in my paper..Steve

---------- Forwarded message ----------
From: [redacted] <[redacted]
Date: Thu, Apr 18, 2013 at 10:30 AM
Subject:
To: [redacted]>

Got a lot done on the analysis finally. Just more specificity around the mutations, etc. No smoking gun, but for the report I now feel I have something more substantive to say that makes a credible analytic statement.

I will have the time tomorrow to write it to completion and I will send it along. I apologize I have just been overwhelmed with email/work/meetings/travel.

In the meantime, an interesting paper on complexity and the origin of life. Basically proposes that given the complexity life began pre-Earth. Support for panspermia.

http://phys.org/news/2013-04-law-life-began-earth.html

But also suggests, that if the ET seeding hypothesis is true, that it must have started billions of years PRIOR since they would have had to come to technological completion at least a few hundred years prior so they could then start spreading things all over the Universe

Best regards,

[redacted]

==

[redacted].

[redacted]

"Without *DEVIATION* from the norm there can be no **Evolution**"

"Absence of evidence is not evidence of absence"

| Director, NHLBI Proteomics Center for Systems Immunology | Baxter Laboratory for Stem Cell Biology

| Department of Microbiology and Immunology | Center for Clinical Science Research

| Stanford School of Medicine | [redacted]

| Medical School Web Profile for Dr. Nolan [redacted]

==